HOMEWORK QUESTIONS

Lecture 3

Q1 - Program to cyclically rotate an array by one in clockwise direction

Input: arr[] = {1, 2, 3, 4, 5}

Output: arr[] = {5, 1, 2, 3, 4}

Lecture 4

Q2 - Check whether two strings are anagram of each other or not

Input1: string a = “abc” string b = “cab”

Output1: “Yes”

Input2: string a = “pqr” string b = “rst”

Output1: “No”

Lecture 6

Q3 - Given an array,

print the majority element if it is present otherwise print “no majority element”.

Majority Element - The majority element is the element that appears more than (n/2) times.

```
Input1: arr = [2,2,1,1,1,2,2]

Output1: 2

Input2: arr = [1,1,5,9]

Output2: “no majority element”
```

Lecture 7

Q4 - You are given an array where arr[i] is the price of a given stock on the ith day.

You want to maximize your profit by choosing a **single day** to buy one stock and choosing a **different day in the future** to sell that stock.

Print the maximum profit you can achieve from this transaction. If you cannot achieve any profit, print 0.

```
Input1: arr = [7,1,5,3,6,4]

Output1: 5

Input2: arr = [7,6,4,3,1]

Output2: 0
```

Lecture 8

Q5 – Print the first non-repeating character in the given string.

Input1: string s = “strings”

Output1: t

Input2: string s = “abcddbca”

Output2: “Not found”